

Newton-Raphson MLE for Cauchy θ — Exam-Ready Short Answers

Q1. What is the algorithm's goal?

Find MLE $\hat{\theta}$ maximizing Cauchy log-likelihood $\ell(\theta)$, by solving score $U(\theta)=0$ via Newton-Raphson (no closed form exists).

Q2. State the model and log-likelihood.

$f(x|\theta) = 1 / [\pi(1+(x-\theta)^2)]$. For n obs:

$$\ell(\theta) = -n \cdot \log(\pi) - \sum \log[1+(x_i-\theta)^2]$$

Q3. Score $U(\theta)$ and Hessian $H(\theta)$?

$$U(\theta) = \ell'(\theta) = \sum [2(x_i-\theta)] / [1+(x_i-\theta)^2]$$

$$H(\theta) = \ell''(\theta) = \sum [2((x_i-\theta)^2-1)] / [1+(x_i-\theta)^2]^2$$

Q4. Newton-Raphson update rule?

$\theta^{(t+1)} = \theta^{(t)} - U(\theta^{(t)}) / H(\theta^{(t)})$ — tangent-parabola jump to the root of U ; quadratic convergence near root.

Q5. Why start at the median, not the mean?

Cauchy has no finite mean $\rightarrow$ sample mean unstable/meaningless; median is robust, near the mode, good NR start.

Q6. Why can Newton-Raphson fail here?

$\ell(\theta)$ can be non-concave with multiple roots of $U(\theta) \rightarrow$ bad start may converge to wrong root, oscillate, or diverge.

Q7. Purpose of the two break conditions in the loop?

- $\text{abs}(\text{hess}) < 1e-10 \rightarrow$ avoid division by ≈ 0 (unstable/huge step).
- $!is.finite(\text{theta.new}) \rightarrow$ stop if update becomes Inf/NaN (overshoot).

Q8. Convergence check used?

Stop when $|\theta_{\text{new}} - \theta| < \text{tol} = 10^{-6}$ (successive iterates barely move — converged).

Q9. What do T, L and supply(T, LL) do?

T = grid of θ (0 to 10, step 0.01); $L = \ell(\theta)$ evaluated on $T \rightarrow$ gives the full log-likelihood curve for plotting.

Q10. What do the plot elements show?

- Black curve = log-likelihood $\ell(\theta)$, single peak.
- Red dashed line = true $\theta = 5$.
- Blue dashed line = MLE $\hat{\theta}$.
- Black dots = NR iteration path (iter), climbing to the peak.

Q11. Actual numeric results ($n = 5000$, seed = 123)?

- Start (median) ≈ 5.0145
- Converges in 3 NR steps: $5.0145 \rightarrow 4.9932 \rightarrow 4.9932 \rightarrow 4.9932$
- MLE ≈ 4.9932 (true $\theta = 5$) — error ≈ 0.007

Q12. Conclusion / interpretation?

NR converges fast (quadratic) from a good start; MLE $\approx$ truth confirms correctness; red & blue lines nearly overlap on plot.